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Project Title:	Circadian Entrainment, Sleep-Wake Regulation and Neurobehavioral Performance During Extended Duration Space Flight		
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Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
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Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Task Description:

The success of human exploration missions outlined in the goals for the Vision for Space Exploration and in the Global Exploration Strategy and Lunar Architecture announcement will depend on the crew's ability to remain alert and vigilant while operating sophisticated equipment and following complex procedures. Unfortunately, the space exploration often involves disruptions to planned sleep-wake schedules and/or non-24 hr sleep-wake schedule during these missions. This can impair circadian entrainment to the subsequent sleep-wake cycle which results in sleep impairment, endocrine disturbance, impaired daytime alertness and neurobehavioral function. Our data suggests that most astronauts would exhibit circadian misalignment in the space flight lighting conditions of <25 lux on the windowless middeck of the space shuttle.
Preliminary data from our laboratory reveal that intermittent bright light exposure is effective in maintaining entrainment of the circadian pacemaker to longer-than-24 h days. Other preliminary results demonstrate that blue light (~460 nm wavelength) is more efficient than white light or green light (~ 555 nm wavelength) of equal photon density in resetting the human circadian pacemaker. Given that bright light pulses might not of practical use on the lunar or Martian habitats — due to time and energy constraints — these results demonstrate the need to develop practical and cost-effective countermeasures for addressing the challenges to the human circadian pacemaker during space missions.

When the current grant was submitted in 2003 we originally proposed to test a countermeasure of "blue-enriched" light to entrain the human circadian pacemaker to a 24.65-h day. Specifically, we proposed to test the following hypotheses: i) that synchronization of the human circadian pacemaker to a 24.65-h day initiated at an adverse phase will not be appropriate in the presence of ~100 lux of white light; ii) that inappropriate circadian synchronization will result in the secretion of the sleep-promoting hormone melatonin during waketime, abnormal somatotropic and corticotropic activity, disturbed sleep, and impaired performance and daytime alertness; and iii) that exposure to "blue-enriched" light during the daytime will establish a normal entrained circadian phase in subjects scheduled to a 24.65-h day.

Seven healthy subjects participated in a 71-day inpatient protocol consisting of 3 baseline days (16:8 h wake:sleep) followed by 60 experimental days (24.65 h, 16.43:8.22 h). Light intensity during the experimental days was ~ 100 lux (n=2) and ~50 lux (n = 5) at 137 cm height in the horizontal angle. Circadian phase was measured by the Dim Light Melatonin Onset (DLMO) measured 6 times under dim light (~ 1.8 lux at 137 cm height in the horizontal angle) and postural control (before and after the experimental days, and on experimental Days 6, 20, 34, & 48) and during several segments on the experimental days. The critical manipulation in the study was the timing of sleep on the first experimental day was shifted 12 h compared to baseline; equivalent to a slam shift. To assess entrainment, phase angle was calculated from the difference between DLMO and scheduled bedtime. The results showed that the average phase angle for subjects on the first 24.65h day was 12.5 + .26 hours for the 50 lux group and 8.96 + .96 hours for the 100 lux group, which is indicative of acute circadian misalignment. The average phase angle on the 6th experimental day was 5.8 + 1.6h for the 50 lux group and 7 + 3.7 h for the 100 lux group, showing that subjects continued to be misaligned. The average phase angle on the 15th experimental day was 1.5 + .46. Thereafter they stabilized. The average phase angle was 2.4 + .31h on experimental day 34 and 2.2 + .26h on experimental day 48 (p>.05). The analysis of the cognitive data under these conditions is ongoing.

It is important to note the following: First, our results showed that crew members would be most vulnerable to the detrimental effects of circadian misalignment during the first two weeks of a mission. This is important, because the duration of short-term mission such as those to the ISS and the moon are approximately two weeks long. Second, the data reveal that light intensities that are similar to the lighting conditions during space flight are insufficient to induce rapid entrainment during a short-term mission. Therefore, we concluded that it was critical to develop countermeasures that would facilitate sleep and circadian adaptation rapidly.

We refocused our research efforts on the circadian and sleep-wake challenges associated with short duration lunar exploration. To do so, we modified our studies during the remainder of this grant period to explore this initial period of vulnerability further and test the effectiveness of our proposed countermeasure (blue enriched light) in short term missions. The progress toward developing the blue light countermeasures is moving forward with the continuing work on the modified 12-day protocol. We have completed 6 subjects in the protocol, have 2 subjects currently in the laboratory and several subjects who are being screened into the protocol.

In the coming year, we plan to complete data collection and the analysis of the physiological and neurobehavioral data. With the planned short-term missions outlined in the Global Exploration Strategy and Lunar Architecture announcement, it is vital to mission success to understand the limits of human performance under space flight conditions and develop countermeasures for ensuring health, productivity and safety of astronauts. With the current project, aimed at evaluating the effectiveness of blue-enriched light for circadian misalignment during planned space missions, we expect to provide NASA with a countermeasure which could be incorporated into the design specifications for planned lunar sorties and in the design of the polar lunar base.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Since the finding that appropriately scheduled light/dark exposure is a powerful means of resetting the human circadian pacemaker, bright light has been used in various clinical settings to induce physiologic adaptation in individuals suffering from circadian rhythm disorders (e.g. night shift workers, people with Advanced and Delayed Sleep Phase Syndromes). We have found that appropriately timed exposure to bright light can produce rapid physiologic adaptation of the circadian pacemaker to a single week of night work and facilitate rapid entrainment to a rotating work schedule, as well as enhance the alertness of night workers during their work shifts. Advanced and Delayed Sleep Phase Syndromes (ASPS and DSPS, respectively) are characterized by a marked difficulty in maintaining appropriate timing of sleep during the desired hours, and there is evidence suggesting that circadian misalignment may underlie the pathophysiology of this condition. We and others have reported data from clinical studies which suggest that evening exposure to bright light or early morning exposure to bright light are successful in the treatment of Advanced Sleep Phase Syndrome or Delayed Sleep Phase Syndrome, respectively. Since the current study is designed to evaluate the effectiveness of blue-enriched light exposure over white light as a countermeasure for circadian misalignment during exploration class mission to mars, it has implications for the development of effective and energy-efficient methods for treatment of circadian rhythm disorders. Optimizing the wavelength of light holds the potential for producing shorter, more efficient light treatment regimens. Shorter treatment regimens would not only increase compliance in clinical populations, but would make light treatment more practical in industrial/work settings.

Task Progress:	In our current NSBRI project, we initially proposed to test a countermeasure of "blue-enriched" light to entrain the human circadian pacemaker to the 24.65-h Martian day. As of this past year seven healthy subjects have successfully completed the 71-day inpatient protocol. Our results showed that crew members would be most vulnerable to the detrimental effects of circadian misalignment during the first two weeks of a mission. This is important, because the duration of short-term mission such as those to the ISS and the moon are approximately two weeks long. Moreover, the data reveal that light intensities that are similar to the lighting conditions during space flight are insufficient to induce rapid entrainment during a short-term mission. Based on these results, we felt that it was critical to develop countermeasures that would facilitate sleep and circadian adaptation rapidly. To do so, we modified our protocol from 71 to 12 days to explore this initial period of vulnerability further and specifically test the effectiveness of our proposed countermeasure (blue enriched light) to hasten circadian adaptation to short term missions and thereby improve sleep during scheduled sleep episodes and improve performance during scheduled wake episodes. The progress toward developing the blue light countermeasures is moving forward with the continuing work on the modified 12-day protocol. We have completed 6 subjects in the protocol, have 2 subjects currently in the laboratory and several subjects who are being screened into the protocol. In the coming year we plan to complete the project and complete the analysis of the physiological and neurobehavioral data.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
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